

Derivative Rules: Product & Quotient Rules with Given Values

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Name: _____

Date: _____

Learning Objectives

- Apply the product rule to find derivatives of products of two functions at a given point
- Apply the quotient rule to find derivatives of quotients of two functions at a given point
- Combine derivative rules (constant multiple, sum/difference, product) to differentiate composite expressions

Problems

1. Let U and V be differentiable functions of x with $U(3) = 5$, $U'(3) = -2$, $V(3) = 4$, and $V'(3) = 3$. Find the derivative of the product UV evaluated at $x = 3$.

2. Using the same values as Problem 1 — $U(3) = 5$, $U'(3) = -2$, $V(3) = 4$, $V'(3) = 3$ — find the derivative of U divided by V , evaluated at $x = 3$.

3. Using the values $U(3) = 5$, $U'(3) = -2$, $V(3) = 4$, $V'(3) = 3$, find the derivative of V divided by U , evaluated at $x = 3$.

4. Suppose U and V are differentiable at $x = 2$ with $U(2) = 3$, $U'(2) = -4$, $V(2) = 1$, and $V'(2) = 2$. Find the derivative of $3U$ minus $2V$ plus UV , evaluated at $x = 2$.

5. Let U and V be differentiable at $x = 1$ with $U(1) = -2$, $U'(1) = 5$, $V(1) = 3$, and $V'(1) = -1$. Find the derivative of $4U$ plus UV , evaluated at $x = 1$.

6. Let U and V be differentiable at $x = 0$ with $U(0) = 6$, $U'(0) = 2$, $V(0) = -3$, and $V'(0) = 4$. Find the derivative of U times V , then find the derivative of U divided by V , both evaluated at $x = 0$. Report both answers.



7. Suppose U and V are differentiable at $x = 5$ with $U(5) = -1$, $U'(5) = 3$, $V(5) = 2$, and $V'(5) = -5$. Find the derivative of $2UV$ minus V divided by U , evaluated at $x = 5$.

8. Let U and V be differentiable at $x = 4$ with $U(4) = 7$, $U'(4) = -3$, $V(4) = 5$, and $V'(4) = 1$. Find the derivative of U squared times V , evaluated at $x = 4$. Hint: treat U squared as U times U and use the product rule twice.

9. Suppose U and V are differentiable at $x = 2$ with $U(2) = 3$, $U'(2) = -4$, $V(2) = 1$, and $V'(2) = 2$. Find the derivative of the expression $5U$ divided by the quantity U minus V , evaluated at $x = 2$.

10. Let U and V be differentiable at $x = 3$ with $U(3) = 2$, $U'(3) = 6$, $V(3) = -4$, and $V'(3) = 3$. Find the derivative of the expression U times V divided by U plus V , evaluated at $x = 3$. This requires both the product rule and the quotient rule.



Derivative Rules: Product & Quotient Rules with Given Values — Answer Key

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Answer Key

1. Answer: 7

- Product Rule: $d/dx[UV] = U \cdot V' + U' \cdot V$
 - Substitute values at $x = 3$: $(5)(3) + (-2)(4)$
 - $= 15 + (-8)$
 - $= 7$
-

2. Answer: 23/16

- Quotient Rule: $d/dx[U/V] = (V \cdot U' - U \cdot V') / V^2$
 - Substitute at $x = 3$: $(4 \cdot (-2) - 5 \cdot 3) / (4^2)$
 - $= (-8 - 15) / 16$
 - $= -23 / 16$
-

3. Answer: 23/25

- Quotient Rule: $d/dx[V/U] = (U \cdot V' - V \cdot U') / U^2$
 - Substitute at $x = 3$: $(5 \cdot 3 - 4 \cdot (-2)) / (5^2)$
 - $= (15 + 8) / 25$
 - $= 23/25$
-

4. Answer: -14

- Differentiate term by term: $3U' - 2V' + (U'V + UV')$
 - Substitute at $x = 2$: $3(-4) - 2(2) + ((-4)(1) + (3)(2))$
 - $= -12 - 4 + (-4 + 6)$
 - $= -12 - 4 + 2 = -14$
-

5. Answer: 19

- Differentiate: $4U' + (U'V + UV')$
 - Substitute at $x = 1$: $4(5) + (5 \cdot 3 + (-2) \cdot (-1))$
 - $= 20 + (15 + 2)$
 - $= 20 + 17 = 37$
 - Wait — recheck: $4(5) = 20$; $U'V = 5 \cdot 3 = 15$; $UV' = (-2)(-1) = 2$; Total = $20 + 15 + 2 = 37$
 - Answer: 37
-

6. Answer: $d/dx[UV] = 18$; $d/dx[U/V] = -20/9$

- Product Rule: $UV' + U'V = (6)(4) + (2)(-3) = 24 - 6 = 18$
 - Quotient Rule: $(V \cdot U' - U \cdot V') / V^2 = ((-3)(2) - (6)(4)) / (-3)^2$
 - $= (-6 - 24) / 9 = -30/9 = -10/3$
 - $d/dx[UV] = 18$ and $d/dx[U/V] = -10/3$
-



7. Answer: -17

- Differentiate $2UV$: $2(U'V + UV') = 2((3)(2) + (-1)(-5)) = 2(6 + 5) = 22$
- Differentiate V/U using quotient rule: $(U \cdot V' - V \cdot U') / U^2$
- $= ((-1)(-5) - (2)(3)) / (-1)^2 = (5 - 6) / 1 = -1$
- Combine: $22 - (-1) = 23$
- Wait — re-evaluate: $d/dx[2UV] = 22$, $d/dx[V/U] = -1$, so $22 - (-1) = 23$
- Answer: 23

8. Answer: -161

- Write $U^2V = (U \cdot U) \cdot V$, then use product rule on $U \cdot U$ first: $d/dx[U^2] = 2U \cdot U'$
- Now apply product rule to $U^2 \cdot V$: $d/dx[U^2V] = (d/dx[U^2]) \cdot V + U^2 \cdot V'$
- $= 2U \cdot U' \cdot V + U^2 \cdot V'$
- Substitute at $x = 4$: $2(7)(-3)(5) + (7^2)(1)$
- $= 2(7)(-3)(5) + 49 = -210 + 49 = -161$

9. Answer: -26/4 = -13/2

- Let numerator = $5U$, denominator = $U - V$
- $d/dx[5U] = 5U'$; $d/dx[U - V] = U' - V'$
- Quotient Rule: $[(U - V) \cdot 5U' - 5U \cdot (U' - V')] / (U - V)^2$
- At $x = 2$: $U - V = 3 - 1 = 2$; $(U - V)^2 = 4$
- Numerator: $(2)(5)(-4) - (5)(3)((-4) - 2) = -40 - (15)(-6) = -40 + 90 = 50$
- Wait — recompute: $(U - V) \cdot 5U' = 2 \cdot 5 \cdot (-4) = -40$; $5U \cdot (U' - V') = 5 \cdot 3 \cdot (-4 - 2) = 15 \cdot (-6) = -90$
- Numerator = $-40 - (-90) = -40 + 90 = 50$; Result = $50/4 = 25/2$
- Answer: $25/2$

10. Answer: 66/4 = 33/2

- Let numerator $f = UV$ and denominator $g = U + V$
- $f' = U'V + UV' = (6)(-4) + (2)(3) = -24 + 6 = -18$
- $g' = U' + V' = 6 + 3 = 9$
- At $x = 3$: $f = UV = (2)(-4) = -8$; $g = U + V = 2 + (-4) = -2$
- Quotient Rule: $(g \cdot f' - f \cdot g') / g^2$
- $= ((-2)(-18) - (-8)(9)) / (-2)^2$
- $= (36 + 72) / 4$
- $= 108 / 4 = 27$

